

James Smith

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Summary

Mechatronics Engineering student at the University Of Waterloo, skilled in **Robotics Operating Systems, embedded software, Computer Vision, and Machine Learning**. With hands-on experience developing real time robotics operating systems with integrated AI. Seeking robotics and software engineering co-op opportunities.

Technical Skills

Languages: Python, C++, C

Software Tools: ROS/ROS2, OpenCV, Free RTOS, STM32 Cube IDE, Docker

AI Tools: NVIDIA IsaacLab, PyTorch, Ultralytics YOLO, RESNET, ONNX

CAD & Simulation: SolidWorks, AutoCAD, Draftsight

Experience

Advanced Manufacturing Intern

May 2026 – August 2026

Multimatic

Richmond Hill, ON

- Lead Software Development for a humanoid robot capable of automating complex manufacturing tasks, and presented development to corporate upper management and company owners.
- Designed and developed a humanoid robotics perception/localization/mapping stack using April Tags, custom YOLO CV models and LiDAR/Point-LIO based SLAM with the ability to navigate through rooms to remembered positions with 10cm tolerance.
- Integrated a custom LoRA finetuned VLM with the humanoid stack creating a chatbot based reasoning interface capable of controlling the physical robot.
- Integrated custom IsaacLab trained RL policies designing a tiered policy runner able to use tracked detections for targeting allowing a humanoid robot to execute RL policies efficiently, accurately and safely.
- Created a Synthetic Dataset Generator and Computer Vision Model training suite, creating YOLO and RESNET based CV models trained for detection and 3D orientation of company parts, completely removing the need for manual annotations while improving model performance.
- Integrated a collaborative robot with a custom Computer Vision system, loading parts from randomized locations into a fixture with less than 1cm tolerance replacing a tedious manual process.

Controls Subsystem Member

September 2025 – May 2026

Waterloo Rocketry

Waterloo, ON

- Aided in the development of roll control canards, working on the embedded software stack (Free RTOS), and the onboard hardware.
- Wrote driver code for an IMU utilizing STM32 Cube IDE to configure a Direct Memory Access channels for automatic I2C reads, eliminating a 500hz CPU task while increasing data acquisition speeds.
- Designed an external camera mounting system for viewing the exterior of the supersonic rocket, optimizing the aerodynamic shape of the parts and camera FOV.

Program Leader

June 2022 – Jan 2026

City Of Oshawa

Oshawa Ontario, ON

- Planned and instructed recreational programs for the municipality achieving over 100% registration rate.
- Supervised a 200,000 square foot facility, addressing safety/patron issues and monitoring program attendance.
- Provided individualized care for special needs youth throughout a summer camp day, adapting programming and desinging behavior management strategies ensuring participant and family satisfaction.
- Utilized adaptive technology such as VOLT hockey carts, wheel chairs, alternative communication devices and AFOs, to adapt recreational programming.
- Worked with safety protocols and handled medication ensuring participant safety throughout the day.

Education

University Of Waterloo

Waterloo, ON

B.S. in Mechatronics Engineering

Expected

Certifications and Awards

Certified Solidworks Associate

December 2025

dassault Systems

Governor General Graduation Award

June 2025

Monsignor Paul Dwyer C.H.S